CS & IT ENGINEERING

Computer Network

IPv4 Address

Lecture No. - 12

By - Abhishek Sir

Recap of Previous Lecture

Topic IP Configuration

Topic ARP

Topics to be Covered

Topic ARP

Topic Supernetting

ABOUT ME

Hello, I'm Abhishek

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[MCQ] IISC [GATE-IT-2008] [2 Mark]

#Q. Host X has IP address $\boxed{192.168.1.97}$ and is connected through two routers R1 and R2 to another host Y with IP address $\boxed{192.168.1.80}$, Router R1 has IP addresses $\boxed{192.168.1.135 \text{ and } 192.168.1.110}$, R2 has IP addresses $\boxed{192.168.1.67 \text{ and } 192.168.1.155}$, the netmask used in the network is 255.255.255.224;

Which IP address should X configure its gateway as?

(A) 192.168.1.67

(B) 192.168.1.110

(C) 192.168.1.135

(D) 192.168.1.155

$\boxed{\text{Ans: B}}$

Netmask = 255 . 255 . 255 . 224 (27 one's)

97 = 01100001
110 = 01101110

135 = 10000111
155 = 10011011

67 = 01000011
80 = 01010000

[MCQ] (Common Data) [GATE-IT-2008] [2 Mark]

#Q. Given the information in previous question, how many distinct subnets are guaranteed to already exist in the network?

A 1

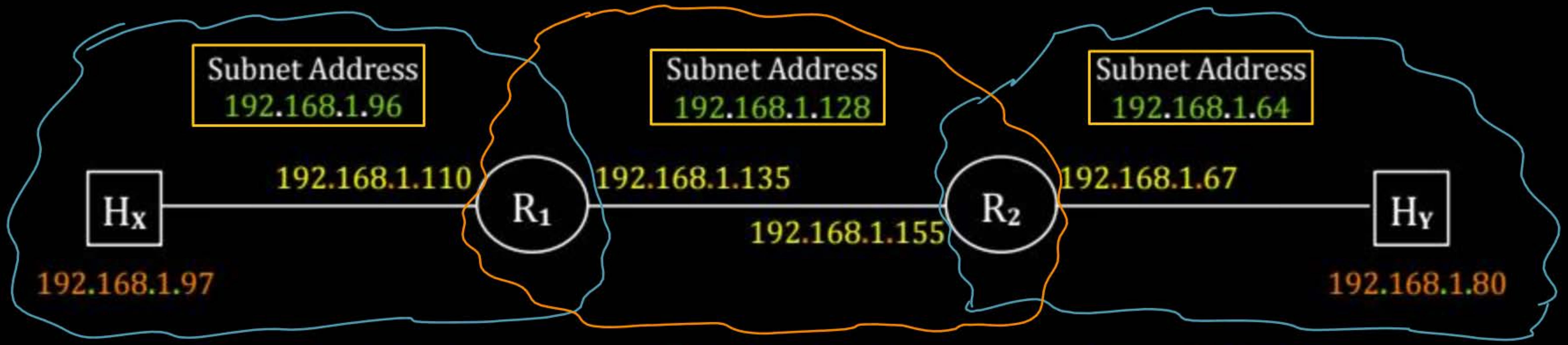
B 2

C 3

D 6

$\boxed{\text{Ans: C}}$

Netmask = 255 . 255 . 255 . 224



97 = 01100001
110 = 01101110

96 = 01100000

135 = 10000111
155 = 10011011

128 = 10000000

67 = 01000011
80 = 01010000

64 = 01000000

[MSQ] (H.W.) (IISC) [GATE-2024][1 Mark]

#Q. Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

A. The destination IP address is the IP address of R

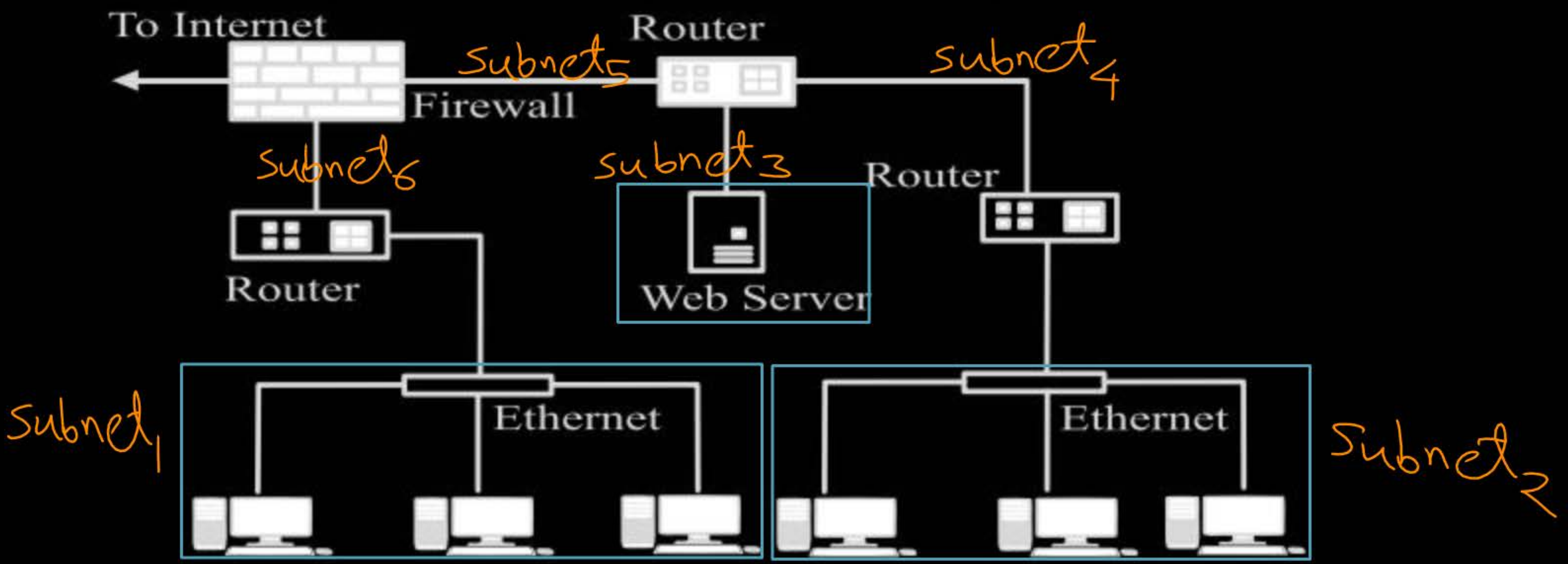
B. The destination IP address is the IP address of Y

C. The destination MAC address is the MAC address of S

D. The destination MAC address is the MAC address of Y

[MCQ] [GATE-2022][1 Mark]

#Q. Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below.



What is the number of subnets inside the enterprise network?

(A) 3

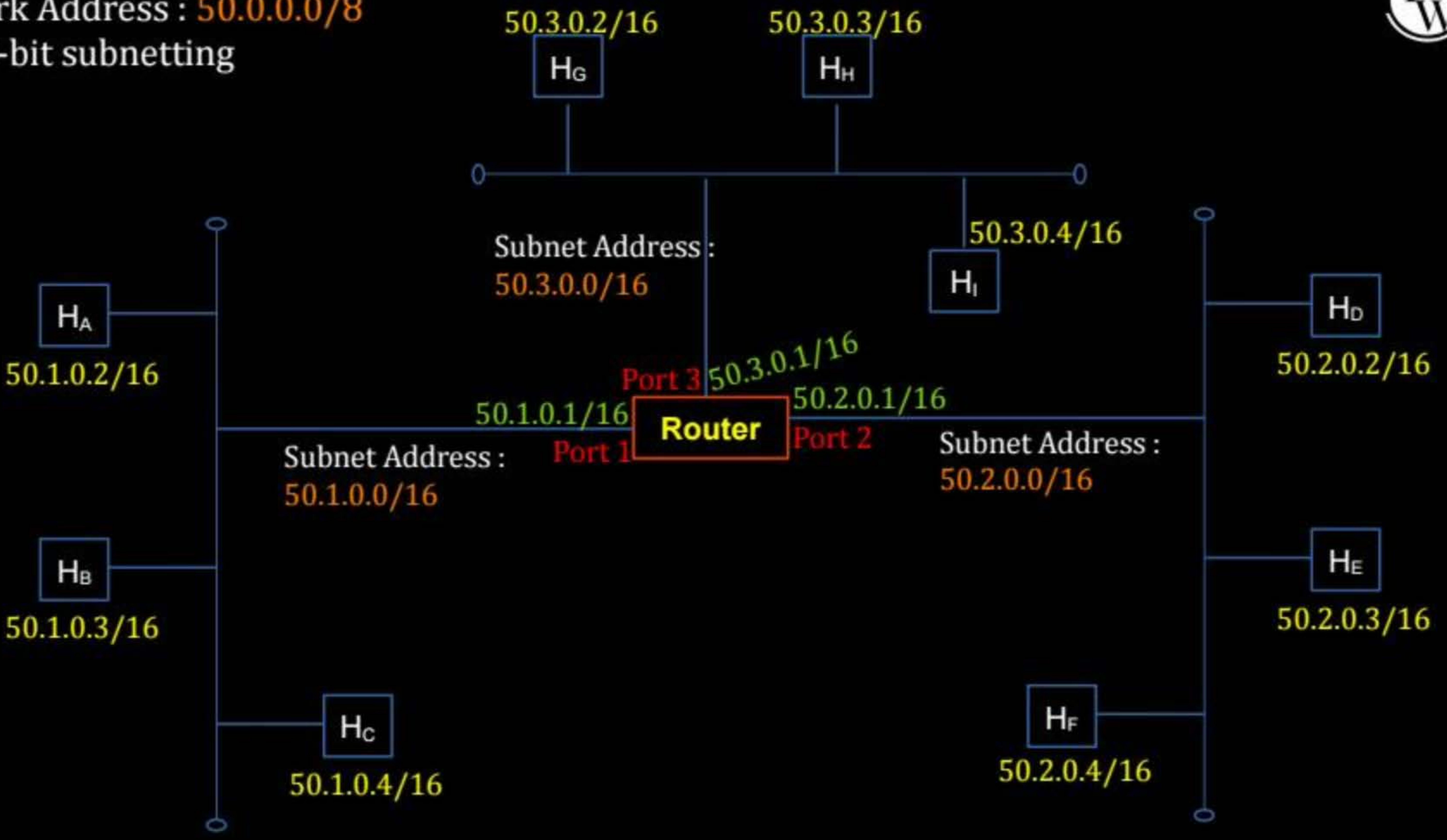
(B) 12

(C) 6

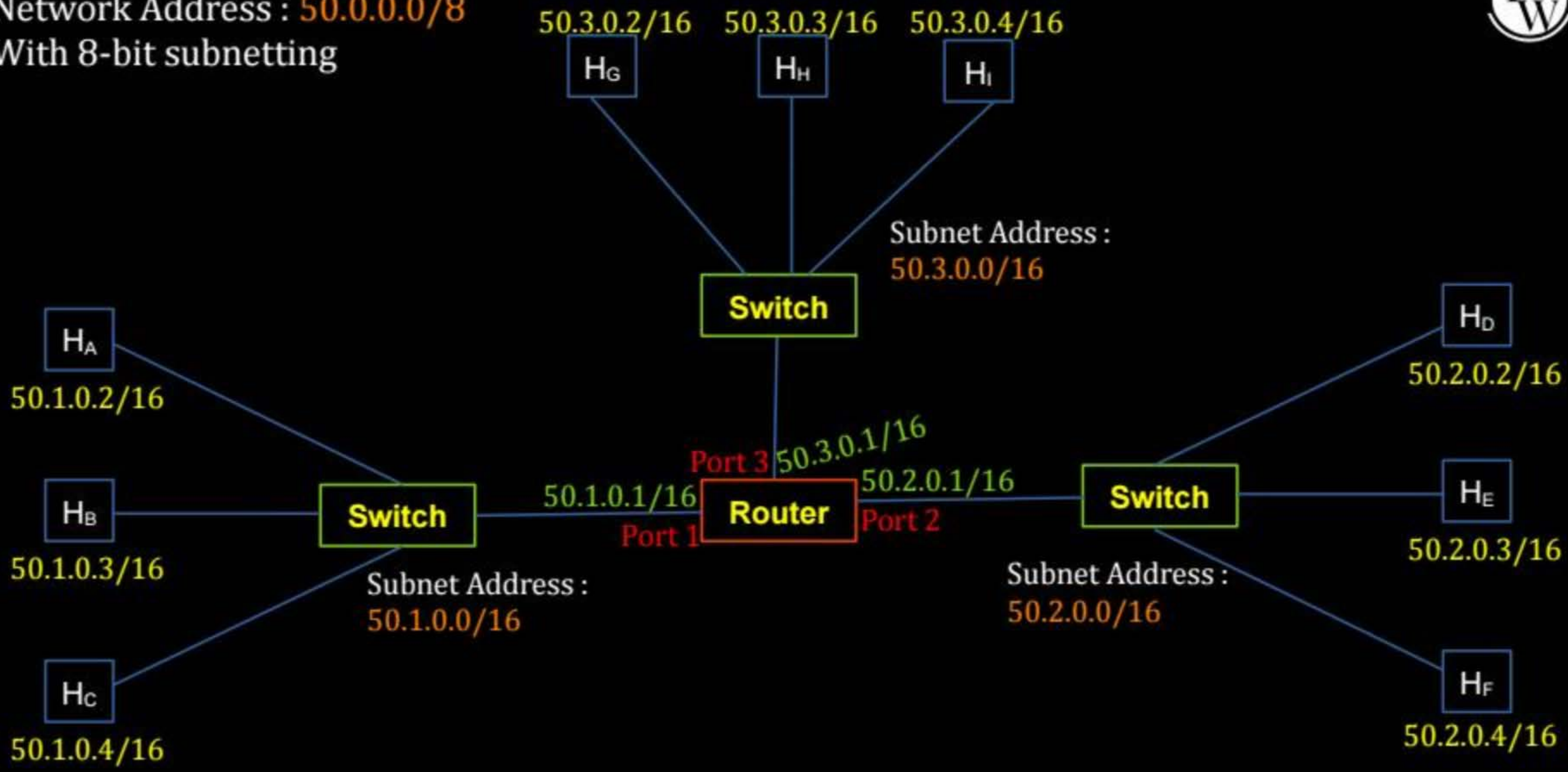
(D) 8

Ans: C

Network Address : 50.0.0.0/8
With 8-bit subnetting



Network Address : 50.0.0.0/8
With 8-bit subnetting



Topic : ARP Packet

[ARP Query ⇒ ARP Reply]
[Same Format]

→ ARP Packet always encapsulated into payload field of $\boxed{\text{“Frame”}}$

→ ARP Query (Request) always broadcast
[Destination MAC address = FF:FF:FF:FF:FF:FF (All 48 bits are one)
Broadcast MAC address]

→ ARP Reply always unicast
[Destination MAC address = MAC address of ~~Host~~ Device who raised ARP query]

[MCQ] H.W. IIT-M [GATE-2019][2 Mark]

#Q. Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

A. X sends an ARP request packet with broadcast IP address in its local subnet

B. X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X

C. X sends an ARP request packet with broadcast MAC address in its local subnet

D. X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X

Topic : ARP Packet

Frame

Destination MAC Address	Source MAC Address	Type (2 Byte)	Payload [ARP Packet] (28 bytes)	FCS [CRC] (4 Byte)

Header

Footer

Hardware Type (2 Byte)		Protocol Type (2 byte)
Hardware Length	Protocol Length	Operation (2 byte)
Source MAC Address (6 byte)		
Source IP Address (4 byte)		
Destination MAC Address (6 byte)		
Destination IP Address (4 byte)		

Topic : ARP Table / ARP cache

→ Used to keep the reply of ARP queries for some time

IP Address	MAC Address	Time to Live
Host B IP	Host B MAC	
255.255.255.255	FF:FF:....FF	Always (∞)
Directed B'cast IP Add.	FF.. FF	Always (∞)

Dynamic

Static

Topic : Loopback Address

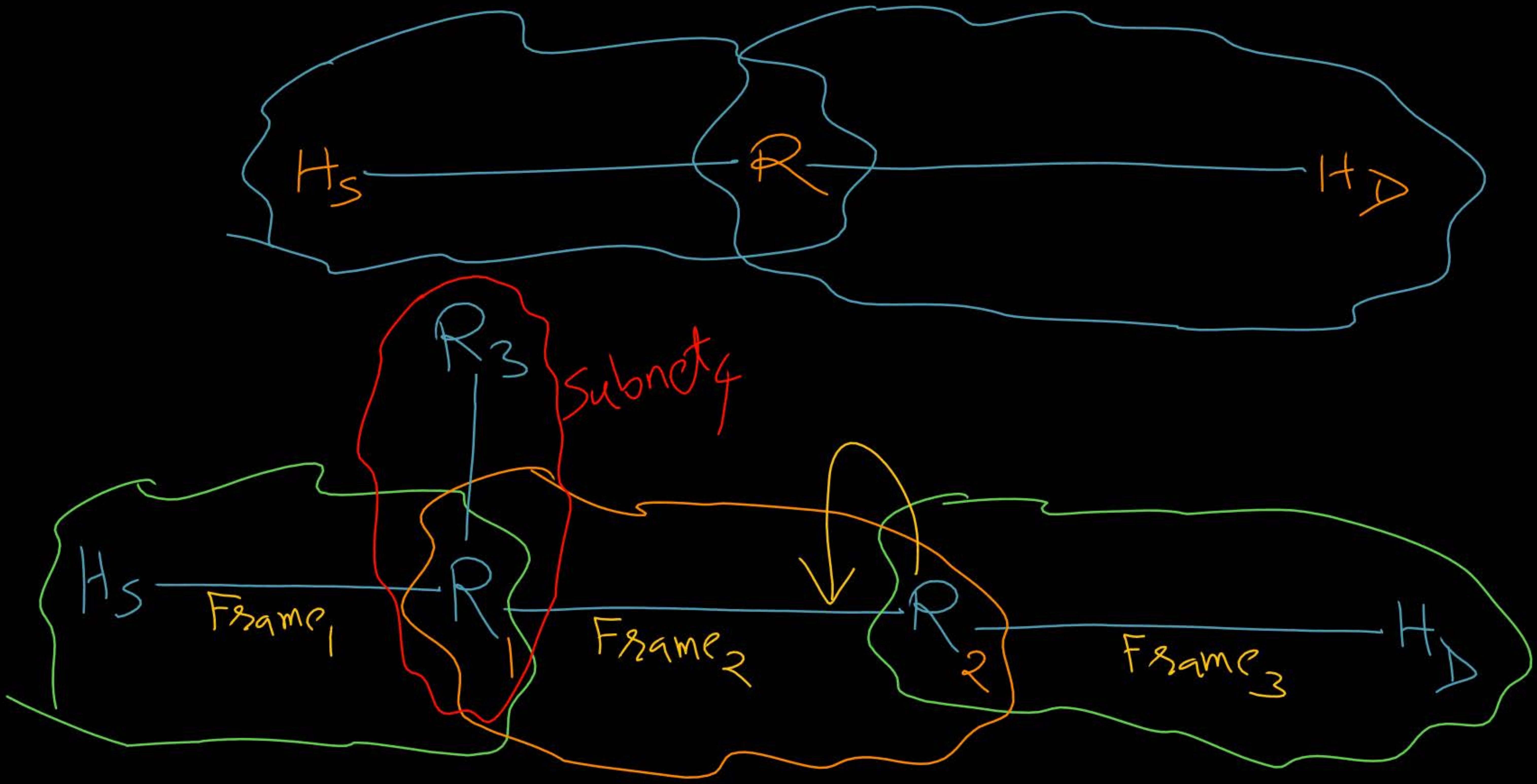
127. Anything

- → Know as "localhost"
- → In IPv4 loopback address can be anyone from [127 . 0 . 0 . 0] to [127 . 255 . 255 . 255]
- → In IPv6 loopback address is "::1" [00. . . . . .001]
- → For diagnostic purpose to verify the internal connectivity

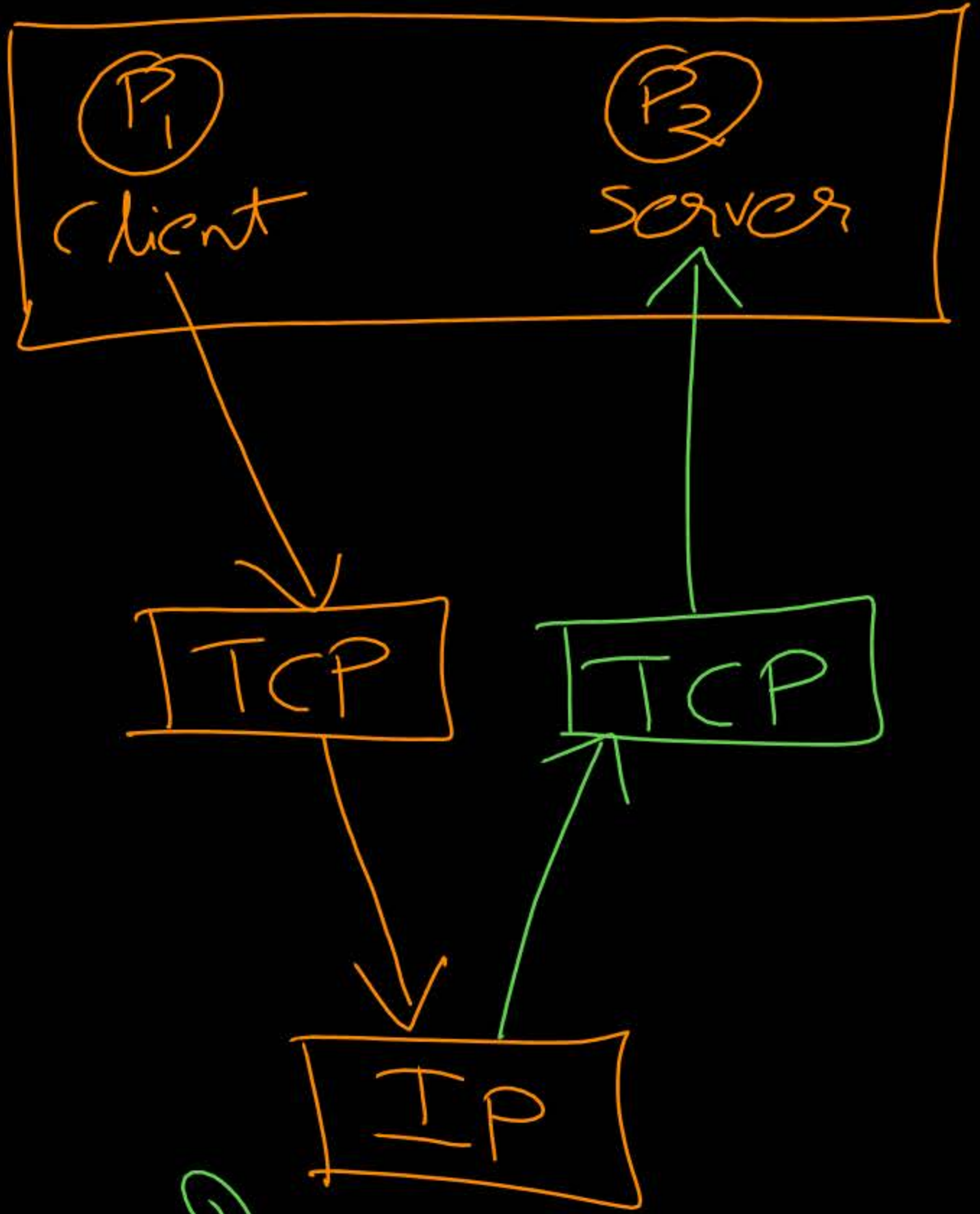
(1) This Host → 0.0.0.0

(2) Limited Broadcast Add → 255.255.255.255

(3) Local Host 127. Anything



Hs
R
HD
R3
Subnet4
Hs
Frame1
R1
Frame2
R2
Frame3
HD



Dest. IP Add.
= local Host

Topic : Supernetting → Reverse process of subnetting

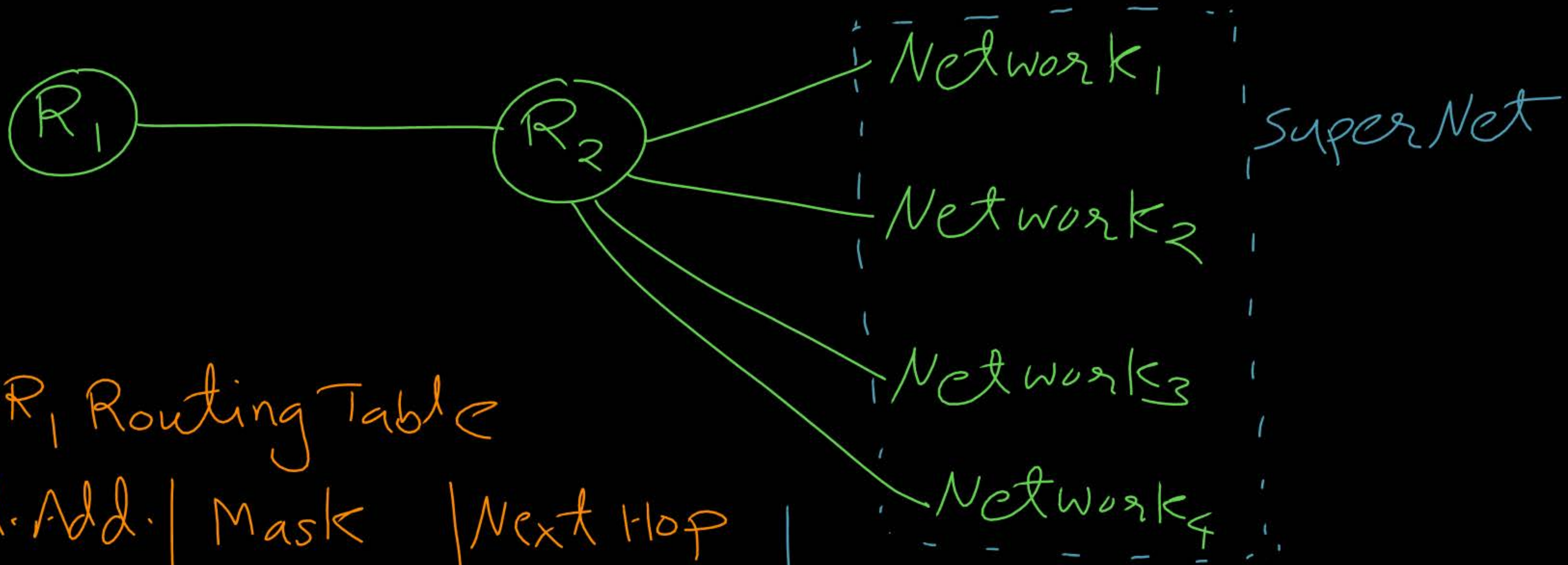
→ Prefix (Route) Aggregation / CIDR aggregation

→ Combining (logically) smaller networks into single large network

→ Allow more efficient routing (Fast Routing)

→ Reduces the number of entries in routing table ✓

→ All the Networks should be contiguous
[Block of addresses having contiguous prefixes]



R1 Routing Table

Net. Add.	Mask	Next Hop
NetAdd1	Maskn	R2
Net Add2	Maskn	R2
Net Add3	Maskn	R2
Net Add4	Maskn	R2

R1 Routing Table

Net Add	Mask	Next Hop
SuperNetAdd	Mask(n-k)	R2

Topic : Supernetting

Example 1 : Suppose network id field size are 6 bits and host id field size are 2 bits. Consider following Network Addresses of networks, what should be the supernet address?

Net. Add.1 : 1 0 1 0 0 0 0 0 / 6

Net. Add.2 : 1 0 1 0 0 1 0 0 / 6

Net. Add.3 : 1 0 1 0 1 0 0 0 / 6

Net. Add.4 : 1 0 1 0 1 1 0 0 / 6

supernet₁ =

1 0 1 0 0 0 0 0 / 5

1 0 1 0 1 0 0 0 / 5

supernet₂ =

Supernet Address : 1 0 1 0 0 0 0 0 / 4

Topic : Supernetting

SuperNet = 10100000/4

supernet1 =

Example 1 : 10100000/5

supernet2 =

10101000/5

Net. Add.1 :	1 0 1 0 0 0 0 0 / 6
	1 0 1 0 0 0 0 1
	1 0 1 0 0 0 1 0
	1 0 1 0 0 0 1 1
Net. Add.2 :	1 0 1 0 0 1 0 0 / 6
	1 0 1 0 0 1 0 1
	1 0 1 0 0 1 1 0
	1 0 1 0 0 1 1 1

Net. Add.3 :	1 0 1 0 1 0 0 0 / 6
	1 0 1 0 1 0 0 1
	1 0 1 0 1 0 1 0
	1 0 1 0 1 0 1 1
Net. Add.4 :	1 0 1 0 1 1 0 0 / 6
	1 0 1 0 1 1 0 1
	1 0 1 0 1 1 1 0
	1 0 1 0 1 1 1 1

Topic : Supernetting

Super Net Add. = 10100000/4

Example 1 :

Net. Add.$_1$: 1 0 1 0 0 0 0 0 / 6	Net. Add.$_3$: 1 0 1 0 1 0 0 0 / 6
1 0 1 0 0 0 0 1	1 0 1 0 1 0 0 1
1 0 1 0 0 0 1 0	1 0 1 0 1 0 1 0
1 0 1 0 0 0 1 1	1 0 1 0 1 0 1 1
Net. Add.$_2$: 1 0 1 0 0 1 0 0 / 6	Net. Add.$_4$: 1 0 1 0 1 1 0 0 / 6
1 0 1 0 0 1 0 1	1 0 1 0 1 1 0 1
1 0 1 0 0 1 1 0	1 0 1 0 1 1 1 0
1 0 1 0 0 1 1 1	1 0 1 0 1 1 1 1

Topic : Supernetting

Example 2 :- Consider following Network Addresses of networks. What should be supernet address ?

H.W.

150 . 125 . 160 . 0 / 23

150 . 125 . 162 . 0 / 23

150 . 125 . 164 . 0 / 23

150 . 125 . 166 . 0 / 23

Supernet Address :

2 mins Summary

Topic ARP

Topic Supernetting

THANK - YOU

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